Semester VII **B. Tech CSE (AI)**

BCSAI 703: ONLINE MACHINE LEARNING

3L+0T+0P+3C **MM:100**

Unit 1: Introduction

Introduction to Bandit Algorithms, Language , Probabilty spaces, Independence, Integration and Expectation, batch to online setting,

Adversarial setting with full information, Halving algorithm, WM Algorithm, Regret Lower Bounds

Unit 2: Stochastic Setting

STOCHASTICS processes, Markov Chains, Martingales, Stopping Times, Regret notions, concentration inequalities, Stochastic Bandit Algorithms, UCB, KL-UCB

Unit 3: Contextual Bandits

Overview, Regret Analysis, Stochastic Linear Bandits, Method of Mixtures, Kiefer-Wolfowitz Theorem, Sparse Stochastic Linear Bandits

Unit 4: Pure Explorations

Fixed confidence, budget, pure exploration setups, Best Arm identification, Bayesian Bandits, Gittins Index, Thompson Sampling

Unit 5: Pure Explorations Algorithms

LUCB,KL-LUCB, lil'UCB, Partial Monitoring, Optimal Policy, Proof of Upper and Lower Bound

Text/Reference Books

1. Bandit Algorithms by Tor Lattimore snd Csaba Szepesvari, Cambridge University Press
2. Bandit Algorithms for Website Optimization by John Myles White,O'Rielly Media
3. Regret Analysis of Stochastics and nonstochastic multi-armed bandit problems by Nicolo Cesa-bianchi and Sebastien Bubeck
4. J.A. Tropp,Foundations and Trends in Machine Learning